Lecture plan · for the lecturer

Data poisoning and backdoors

AI security · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Classify poisoning attacks.
- Explain backdoors.
- Defend the data supply chain.

Activity: Map the supply chain

1. Warm-up (5 min): Tell the Tay story and ask how the model was changed without touching its code.
2. Map (15 min): Groups draw a spam filter's data supply chain from reports to retraining.
3. Attack (15 min): Mark where an attacker could insert data and what goal each serves.
4. Control (20 min): Propose one control per entry point and say what it cannot stop.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which entry point was easiest?
- Would clean accuracy reveal it?
- Which control covers most?

Exit ticket

- Name one dataset you use whose provenance you cannot verify.

Try it in your next lecture
Give groups a spam filter retrained weekly on user reports; each group marks every point where an attacker could insert data and proposes one control for each.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Training on scraped web images. Which step best protects against poisoning?**

◯ More scraped images ◯ Provenance tracking, filtering and a trusted test set

◯ A bigger model

2 **Size the attack.**

A dataset has 20,000 images. Poisoning 1% means _____ images; poisoning 0.5% means _____.

Word bank: 20 · 100 · 200 · 2000

3 **Which help defend a model trained on scraped data? Choose all that apply.**

☐ Recording and verifying data provenance ☐ Relying on high clean accuracy

☐ A trusted held-out test set ☐ Outlier and spectral-signature filtering

4 **A model with 98% clean test accuracy cannot contain a backdoor.**

◯ True ◯ False

5 **Match each attack to its effect.**

Availability poisoning	• • • •	Failure only when a trigger appears
Targeted poisoning	• • • •	A crafted input at test time
Backdoor	• • • •	Lower accuracy on all inputs
Evasion	• • • •	Failure on chosen inputs

Answers · for the lecturer

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1 **Provenance tracking, filtering and a trusted test set**

Control and audit what enters the data.

2 **A dataset has 20,000 images. Poisoning 1% means 200 images; poisoning 0.5% means 100.**

0.01 × 20,000 = 200 and 0.005 × 20,000 = 100: a backdoor needs very few rows.

3 **Recording and verifying data provenance; Outlier and spectral-signature filtering; A trusted held-out test set**

Backdoored models keep high clean accuracy.

4 **False**

Backdoors are designed to leave clean accuracy unchanged.

5 **Availability poisoning — Lower accuracy on all inputs**
Targeted poisoning — Failure on chosen inputs
Backdoor — Failure only when a trigger appears
Evasion — A crafted input at test time

Poisoning acts in training; evasion acts at test time.

✎ **For you to think about**

Where do the datasets in your course come from, and has any student ever asked whether they can be trusted?